Reliable Annotation and Evaluation of Malaysian Malay-English Code-Mixed Sentiment Data: A Critical Review of an Unresolved Problem

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Abstract

Malaysian social media users mix English and Malay inside single messages, producing the contact variety known locally as Manglish. The studies reviewed here usually treat sentiment analysis of Malay-English mixed writing as a modelling problem. This review indicates that the more limiting constraint is earlier in the process, in the quality of the sentiment labels and in the evidence offered that those labels are sound. The review covers annotation practice and reliability evaluation for Malaysian Malay-English code-mixed sentiment data. Comparative benchmarking of sentiment classifiers is outside its scope. Three contributions follow. First, the annotation strategies used for low-resource code-mixed text are organised into three tiers, and the trade-off between label fidelity and scalability is examined across them. Second, two sources of neutral-label skew are separated, one produced by the labelling tools and the other by genuine uncertainty among annotators. The four Malay-English sentiment resources are then examined against what a reliability claim requires, and none supplies it. Third, the class imbalance in the label distribution is shown to be the condition that makes Cohen's kappa more difficult to interpret. The review closes with a research agenda covering minimum reporting standards, agreement measurement under class imbalance, a shared annotation convention for affect-bearing particles, and the untested assumption that annotators of different types make different errors.

Keywords: *sentiment analysis, code-mixed text, Manglish, annotation reliability, inter-annotator agreement*

1. Introduction

Sentiment analysis, also known as opinion mining, is the automated task of deciding whether a piece of text expresses a positive, negative or neutral opinion (Poria et al., 2023). Organisations use it to measure public opinion at scale, in domains that include consumer products and services, healthcare, political elections and financial services (Nazir et al., 2022). However, the tools used for this task assume that their input is written in one standard language. Joshi et al. (2020) highlight that only a small number of the world's languages are represented in current language technologies, with lexicons, pre-trained models and benchmark corpora concentrated in those few languages. Among these three, a pre-trained model is one that has already learned general patterns from a large body of text, and is then adapted to a task such as sentiment analysis (Devlin et al., 2019). Many of these language resources are developed primarily for individual languages, so text

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in which two languages alternate within a single message does not meet the conditions for which they were designed. Doğruöz et al. (2021) call this alternation code-switching, and report that its effect on model performance varies by task. In their survey, current models perform well on code-switched language identification and part-of-speech tagging, but only slightly better than chance on natural language inference. Winata et al. (2023), surveying a wider set of studies, also report that performance on code-switched text is often lower than performance on monolingual text for the same task, though not uniformly so. Code-switching between English and Malay is common in Malaysian social media text (AlAfnan et al., 2025).

Malaysia is a multilingual and multi-ethnic country. Drawing on Ethnologue figures, Coluzzi (2017) records around 140 languages in use in the country, and reports that code-mixing between a speaker's first languages and English or Malay is widespread in everyday communication. AlAfnan et al. (2025) report that English and Malay are among the languages most frequently combined in Malaysian written communication. The same study describes social media users alternating between the two languages within a single message rather than writing in one language at a time. Two local terms are used for this code-mixed writing, and they refer to different combinations of languages. AlAfnan et al. (2025) describe *Manglish* as a hybrid form that combines Malay and English, whereas Romadhona et al. (2022) describe *Bahasa Rojak* as a variety that combines English, Malay and Chinese. This review is concerned with Malay-English mixing. *Manglish* names that combination, while *Bahasa Rojak* names a broader one that also includes Chinese.

Manglish is not always described in the same terms. AlAfnan et al. (2025) note that it is sometimes viewed as having a negative effect on standard English and Malay rather than as a legitimate language variety in its own right. The structure of *Manglish*, however, is systematic rather than random. Treffers-Daller et al. (2022) analyse the classroom speech of two Malaysian teachers of English and find that Malay-English switching there follows describable syntactic patterns. AlAfnan et al. (2025) report that speakers use *Manglish* for identifiable purposes, which include communicating efficiently, marking identity and building rapport.

The same systematicity appears at the level of spelling. Maskat and Rahman (2020) document the Malay negator *tidak* being written as *x*, and *tidak ada* being written as *takde*, *xde* or *tade*. They sort this variation into five categories and map two of them back to their standard forms, and this categorisation suggests that the variation follows recurring patterns rather than occurring arbitrarily. A dictionary of standard Malay, such as *Kamus Dewan*, does not list these non-standard forms (Dewan Bahasa dan Pustaka, 2005). Tools that rely on such a dictionary may therefore treat them as out-of-vocabulary items, meaning words absent from the reference lexicon (Maskat and Rahman, 2020). Abu Bakar et al. (2020) describe Malay social media text as misspelled, out-of-vocabulary, ill-formed and noisy. This review reads that description as a statement about the reference standard being applied rather than about the care taken by the writer. These writing conventions are consistent within the observed variation, but they differ from the conventions that standard natural language processing tools were designed to handle.

These differences between Malaysian writing conventions and the assumptions built into standard tools raise a question about the source of the difficulty. When a

sentiment model performs poorly on Malay-English mixed text, one common explanation concerns the model architecture (Doğruöz et al., 2021; Winata et al., 2023). Under this view the model processes code-switched input less effectively than monolingual input, so a larger model, or one further trained on Malay-English or other Southeast Asian data, may perform better. The evidence for this explanation is mixed. Khanuja et al. (2020) report that multilingual models fine-tuned on code-switched data achieve the strongest scores among the systems evaluated on their benchmark. Adilazuarda et al. (2022) find that robustness to code-mixing varies with the embedded language, and attribute the variation to bias in the pre-training corpus. In contrast, Winata et al. (2021) find that pre-trained multilingual models do not necessarily produce high-quality representations of code-switched text. Conneau et al. (2020) describe a trade-off between two effects in multilingual models. Adding languages can improve performance on each of them, an effect they call positive transfer. Beyond a point, however, the modelling capacity available to any individual language falls, an effect they call capacity dilution, and the quality of its representations may decline. Doğruöz et al. (2021) identify a further limitation that is not architectural: the multilingual models in common use are trained mainly on monolingual corpora and may receive little code-switched text during pre-training. Model choice therefore accounts for part of the difficulty but not all of it.

Another source of difficulty arises one step earlier, at the point where sentiment labels are created. Supervised sentiment analysis requires a sentiment label for every message before a model can be trained or evaluated on it. The amount of labelled data available therefore sets a bound on what can be trained and on what can be tested. Poria et al. (2023) describe this reliance on annotated data as often limiting, and note that it has renewed interest in supplying models with external knowledge such as sentiment lexicons. A sentiment label is assigned either by a human annotator or by an automatic labelling method, and this step is called annotation. The labels form the supervision signal used to train a model, and the labels in a test set are the standard against which the model's predictions are compared. Incorrect labels can affect both model training and subsequent evaluation. During training, a model may learn the patterns associated with incorrect labels, which can reduce predictive accuracy (Frénay and Verleysen, 2014). During evaluation, incorrect labels in a test set can bias the measurement of model performance. Northcutt et al. (2021) show that label errors can change which of two models is ranked higher on a benchmark. Bowman and Dahl (2021) name annotation reliability as one of the conditions a sound evaluation benchmark must meet. An incorrect label can therefore have an effect at two separate stages, first when the model is trained and again when its performance is measured.

Those two stages apply to supervised learning in general. For Malaysian Malay-English code-mixed sentiment data in particular, the question is whether the model or the sentiment labels are the greater constraint at present. Three findings point to the labels, and each is developed in the sections that follow. First, this review identifies four Malay-English sentiment resources (Kasmuri and Basiron, 2019; Romadhona et al., 2022; Kong et al., 2023; Mohd Shamsuddin et al., 2024). None of them meets all four of the criteria a general-purpose resource would need. Those criteria are (1) at least ten thousand labelled messages; (2) a label distribution that is not dominated by a single sentiment class; (3) text drawn from everyday social

media posts rather than from blogs or reviews; and (4) an assessment of annotation reliability by human annotators. Each of them satisfies some of these criteria but not all of them. Second, the largest of them is labelled 78.23% neutral (Mohd Shamsuddin et al., 2024). That skew towards the neutral label appears to originate in the labelling method rather than in Malaysian discourse. Third, the studies introducing two of the four resources report no inter-annotator agreement coefficient at all (Kong et al., 2023; Mohd Shamsuddin et al., 2024). Where a coefficient is reported, it was measured on a two-way judgement rather than on the three-way positive, negative and neutral labelling for which the resource is used (Kasmuri and Basiron, 2019; Romadhona et al., 2022).

The second and third findings are closely related rather than independent. Reliability is normally assessed by having independent annotators label the same text and computing a chance-corrected agreement coefficient, such as Cohen's kappa (κ) (Cohen, 1960; Artstein and Poesio, 2008). Cohen's κ becomes difficult to interpret when one category dominates the label distribution, because the agreement expected by chance rises with that skew (Feinstein and Cicchetti, 1990; Byrt et al., 1993). Di Eugenio and Glass (2004) call this effect the prevalence problem. The skew towards the neutral label described in the second finding can therefore make Cohen's κ, the statistic named in the third, more difficult to interpret. Neither correction would be sufficient without the other. These four resources are therefore limited both in what they contain and in what is known about the reliability of their labels. That shortage is not confined to them, since Selvaraju et al. (2024) report that comprehensive datasets and annotated resources for code-mixed Malay text are still lacking more generally.

These three findings lead to the question this review examines: why reliable annotation and evaluation of Malaysian Malay-English code-mixed sentiment data have not yet been established. Linguistic description, model architecture and application domain are discussed only where they are relevant to that question. Two related topics are not examined here. The first is comparative benchmarking of sentiment classifiers, and the second is sentiment analysis as a general field. Both have already been surveyed in detail (Poria et al., 2023; Nazir et al., 2022). This review does not propose, implement or evaluate a system, and it does not report the authors' own experimental work. Its contribution is an analysis of why that question remains open, together with a set of directions for future research.

This review makes three contributions:

1. It organises the annotation strategies used for low-resource code-mixed text into three tiers, which are manual, semi-automated and fully automated labelling, and examines the trade-off between label fidelity and scalability across them.
2. It distinguishes two sources of neutral-label skew, one produced by the labelling tools themselves and the other by genuine uncertainty among annotators. It also examines what the studies that introduced the four Malay-English sentiment resources report about how reliably each was annotated.

3. It shows how the linguistic, annotation and evaluation challenges compound one another rather than merely co-occurring, and sets out a research agenda that follows from that interaction.

The literature examined here was identified through searches of the ACL Anthology, IEEE Xplore, Scopus and arXiv, combining terms for the language variety, the sentiment task and annotation quality. Studies that only benchmark classifiers, with no annotation or reliability component, were not included.

The rest of this paper is organised as follows. Section 2 explains why Malaysian code-mixed writing is difficult to annotate. Section 3 examines annotation practice and its failure modes. Section 4 discusses how annotation reliability is evaluated. Section 5 brings these findings together and sets out directions for future research, and Section 6 concludes the review.

2. Why Malaysian Code-Mixed Text Is Difficult to Annotate

Section 1 identified the sentiment labels, rather than the model, as the greater constraint on sentiment analysis of Malaysian Malay-English text. Three properties of the text explain why those labels are difficult to produce. First, switching between Malay and English occurs inside the sentence. Second, the affect of a message, meaning the attitude it conveys, can be carried by Malay particles that no standard sentiment lexicon lists. Third, phonetic spelling gives a single word many written forms. All three properties are described in the sociolinguistic literature on Malay-English contact (Muysken, 2013; Treffers-Daller et al., 2022; AlAfnan et al., 2025). Each is presented below with its consequence for the annotator, who may be a person or an automatic labelling method.

2.1 Switching occurs within the sentence

Treffers-Daller et al. (2022) analysed 1,044 switched utterances from the classroom speech of two Malaysian teachers of English. In their data, the majority of the switches occurred within the clause rather than between clauses. Within-clause switches took two main forms: a Malay word or phrase inserted into an English stretch, or words from both languages combined in a single shared word order. The authors attribute this pattern to the structure of the two languages rather than to the style of the speakers. English and Malay both place the subject before the verb and the verb before the object, and Malay words change little in form to mark features such as tense. A speaker can therefore move from one language to the other in the middle of a clause with little grammatical adjustment (Treffers-Daller et al., 2022; Muysken, 2013).

These proportions were measured on spoken classroom data and should not be taken as the proportions in social media text. The explanation given by the authors, however, concerns the structure of the two languages and not the medium in which they are used. The same type of switching can therefore be expected in writing. Mohd Shamsuddin et al. (2024) give a written example in the post "Does institut kanser offer free medication cheap chemotherapy", where the Malay noun phrase *institut kanser* is placed inside an English question.

The consequence for annotation is that the boundary between the two languages does not show where the sentiment in a message is located. In monolingual annotation, locating the words that express an opinion is a separate step from judging their polarity. Aspect-based sentiment analysis is one example, in which a sentiment label is assigned to each entity mentioned in a text rather than to the text as a whole (Nazir et al., 2022). The annotator first marks the words that express the opinion and then assigns a polarity to them (Pontiki et al., 2016). In Malay-English mixed text, deciding which words to judge already requires interpretation. Two annotators may therefore assign different labels because they are judging different parts of the message, not because they judge the sentiment differently.

2.2 Affect is carried by particles that belong to neither language

A second difficulty arises once the opinion has been located. The word that carries the affect may be a Malay particle attached to a clause that is otherwise entirely English, a construction Muysken (2013) terms back-flagging. In the data of Treffers-Daller et al. (2022), back-flagging accounts for 11.6% of the switched utterances, and the particle involved most frequently is *lah*. The same study reports that *lah* can attach to a clause that is entirely English, entirely Malay or mixed. Two examples it reproduces from earlier descriptive work are "Buy this lah" and "you understand lah kan".

These particles change the tone of a sentence rather than its factual content. AlAfnan et al. (2025) classify particles as a lexical feature that distinguishes Manglish from standard English and standard Malay. In their account, particles add emphasis, signal politeness or informality, and soften instructions. Their example is "Don't disturb la, he's busy", in which the particle makes the instruction less direct. Treffers-Daller et al. (2022) report the same effect for *lah*, which can shift an utterance from a highly assertive tone to a polite request or a friendly encouragement. In both accounts, the particle leaves what the sentence states unchanged and alters only the attitude it conveys. That attitude is what a sentiment label is meant to record.

Neither source measures how frequently these particles occur in Malaysian social media writing. Treffers-Daller et al. (2022) analyse recorded classroom teaching, and AlAfnan et al. (2025) analyse scripted television dialogue. Both establish that the construction recurs and that it carries affect.

The consequence for annotation is that the particle shows the tone of a message but does not determine its polarity. "Don't disturb la, he's busy" can be read as irritation, which is negative, or as politeness, which is neutral. Treffers-Daller et al. (2022) note the same difficulty for discourse markers in general, whose specific meanings are often difficult to capture even where their pragmatic functions can be evaluated. Two annotators may read the same particle differently, and both may be reading it competently.

Disagreement that arises from more than one justifiable reading is not removed by adding annotators. Pavlick and Kwiatkowski (2019) find that where an item can be read in more than one justifiable way, the disagreement is reproducible. Further annotators divide in the same way rather than converging on one label. A corpus

built by majority vote over such items records the majority reading as though it were the only justifiable one. An agreement coefficient computed over such items therefore reflects genuine differences in interpretation rather than annotator error alone.

2.3 Phonetic spelling gives a single word many written forms

The third property concerns whether the words that carry the sentiment can be recognised at all. Maskat and Rahman (2020) document Malay social media users spelling words as they are pronounced rather than as a dictionary lists them. A single word therefore appears in many written forms. From a corpus of 6,241 Twitter posts, they sort this variation into five categories: spelling variants, Malay-English mixed sentences, loan words spelled according to Malay pronunciation, slang-based spellings, and vowel-less words. Among the spelling variants, *macam mana* appears as *mcmana*, *camne*, *camner*, *mcmner* and *cane*. In the vowel-less category, a word is reduced to its consonants or to a single character, so that *jangan* is written *jgn* and *pada* is written *pd*.

Whether this variation can be reversed automatically was tested by Maskat and Rahman (2020) for two of the five categories, spelling variants and vowel-less words. They used two measures of how closely two spellings resemble one another, n-gram similarity and Levenshtein distance, to map each non-standard form to its standard spelling. The benchmark was a set of 1,176 word pairs identified by hand by a native Malay speaker. The highest precision reported for any configuration was 0.43, meaning that fewer than half of the matches proposed were correct. The authors conclude that similarity matching can detect the two categories but that a more sophisticated technique is needed, and they leave the remaining three categories to future research.

One limit on this evidence matters for this review. English words were removed before the experiment and code-mixed posts were excluded from the evaluation subsets. The reported precision was therefore measured on Malay text with the English taken out, not on Malay-English mixed text. No equivalent measurement for mixed text was found among the studies reviewed here.

The consequence for annotation is that human and automatic annotators differ in what they can recognise. In the study just described, a single native speaker identified the 1,176 word pairs reliably enough for them to serve as the ground truth. Automatic matching on the same ground truth reached a precision of 0.43. A human reader recognises *x* as *tidak*; the automatic procedure largely did not. The two types of annotator therefore make errors on different inputs, and the choice of annotator determines not only cost and speed but which errors a corpus will contain.

3. Annotation Practice and Its Failure Modes

Section 2 described what makes Malaysian code-mixed text difficult to label. This section examines what has been done about it. The annotation strategies reported in the reviewed studies are set out first, with the trade-off each one accepts. The skew towards the neutral label that those strategies produce is examined next.

The section closes with the practice of pooling several annotators, on which the largest Malaysian corpus provides direct evidence.

3.1 Annotation strategies and the fidelity-scalability trade-off

For low-resource code-mixed text, label fidelity and scale are in tension: a corpus can be labelled carefully or it can be labelled in volume. The strategies reported in the reviewed studies differ in which of the two they sacrifice, and they fall into the three tiers summarised in Table 1.

The first tier is manual annotation. Two of the four Malay-English sentiment resources were built this way, and they are the only two whose labels come from human judgement. Kasmuri and Basiron (2019) retained 2,316 sentences from 5,849 drawn from personal blogs, and report $\kappa = 0.83$ for the judgement of whether a sentence was opinionated or factual. Kong et al. (2023) annotated 11,568 of the 108,246 tweets they collected into three sentiment classes, using three independent annotators and a written guideline. Both resources illustrate the cost of choosing fidelity rather than a defect in the method. The labour that produces reliable labels is also what limits how far a manually annotated corpus can grow.

The second tier is semi-automated annotation, which makes the opposite choice: existing tools label at volume, and the limitations of those tools are accepted as the price. Mohd Shamsuddin et al. (2024) built MESocSentiment this way. They collected 229,566 tweets carrying the "#Malaysia" keyword, and after pre-processing and automatic language identification 20,226 code-switched posts remained. Sentiment labels were then assigned by majority vote among three existing sentiment analysers. The authors state that the language identification libraries "reduced the need to find fluent speakers in Malay and English to classify the tweets". The reasoning is justifiable, and on other labelling tasks automated annotators have matched or exceeded human crowd workers (Gilardi et al., 2023). What the tier concedes is that the labels can be no more competent than the tools that produce them.

The third tier replaces those tools with generative language models used directly as annotators. A generative model is not restricted to a fixed vocabulary. It can therefore, in principle, interpret an unfamiliar spelling or slang term from its context, which is the capability the second tier lacks. The peer-reviewed evidence does not yet support that promise. Jadhav et al. (2025) evaluated GPT-4o and Llama 3.1 405B as annotators for Marathi, a low-resource Indian language, across four classification tasks including sentiment analysis. Both scored below a smaller pre-trained model, BERT (Devlin et al., 2019), that had been fine-tuned on each task, by average accuracy margins of 10.2% and 14.1%. The authors conclude that fine-tuned models remain necessary where linguistic resources are scarce. No study examined here evaluates generative annotation on Malaysian code-mixed text, so this tier is the least evidenced of the three.

Table 1. Annotation Strategies for Malaysian Malay-English Code-Mixed Sentiment Data

Tier	Regional example	Scale achieved	Evidence of fidelity	What is conceded
Manual	Kasmuri and Basiron (2019)	2,316 sentences retained from 5,849 candidates, drawn from 23 blogs	$\kappa = 0.83$, for a binary opinionated-or-factual judgement	Scale
Manual	Kong et al. (2023)	11,568 tweets annotated from 108,246 collected	Three annotators and a written guideline; no coefficient reported	Scale
Semi-automated	Mohd Shamsuddin et al. (2024)	19,714 posts from 229,566 collected	None reported; the pipeline has no human annotation stage	Demonstrable fidelity
Fully automated, generative	None published for Malay-English	—	—	Reliability precisely where linguistic resources are scarce (Jadhav et al., 2025)

3.2 Two sources of neutral-label skew

The clearest evidence of what the second tier costs is the class distribution of the corpus it produced. MESocSentiment is labelled 78.23% neutral, 16.33% positive and 5.44% negative across 19,714 posts (Mohd Shamsuddin et al., 2024). Two reasons support interpreting that distribution as an artefact of the labelling method rather than as a property of Malaysian discourse. The first is that the corpus was collected under the "#Malaysia" keyword across sports, politics and the arts, topics that would be expected to attract opinion rather than neutral comment. The second is that a comparable corpus of Malaysian tweets labelled by human annotators is only 23.6% neutral (Kong et al., 2023). Two sources contribute to the skew.

The first source is a scoring function that finds nothing to score. Lexicon-based sentiment tools match each word or phrase in a text against a sentiment lexicon and add the scores of the matched words into an overall score (Selvaraju et al., 2024). That score is then divided into classes by threshold. VADER, one of the three analysers used to build MESocSentiment, returns neutral for any score between −0.05 and +0.05 on a scale from −1 to 1 (Hutto and Gilbert, 2014). It follows that a text whose affect-bearing words are absent from the lexicon contributes little or nothing to the score and is returned as neutral. Under this view, neutral is not always a judgement that a text is affectively balanced. It can also be the default output when nothing was recognised.

The corpus authors' own error analysis indicates that this default output occurs in their data. Mohd Shamsuddin et al. (2024) present six examples comparing the ensemble's label with the correct label. All three misclassified examples are neutral labels that should have been positive, and in all three the sentiment is carried by Malay: *memang cantik sangat*, *selamat deepavali* and *raih gelar juara*. In the correctly classified negative example, the sentiment is carried by an English word, *worthless*. Six author-selected examples give no error rate, but their pattern is

consistent. Where the ensemble succeeded, the affect was in a language its tools covered; where it failed, the affect was not.

The second source is genuine uncertainty among annotators, and it has nothing to do with the tools. Kasmuri and Basiron (2019) note that low agreement in sentiment annotation arises because sentiment is subjective, so judgements differ from one annotator to another and are shaped by each annotator's background. Kong et al. (2023) make the same point about their own task and give it as the reason a clear guideline is needed. As Section 2.2 showed, disagreement that arises from subjectivity is not removed by collecting more judgements.

Separating the two sources matters because they produce the same label and require opposite responses. A genuinely neutral post belongs in the corpus, whereas a falsely neutral post is an error that a better-resourced annotator would not have made. A pipeline built from tools that do not recognise the affect has no way to tell the two apart. A false neutral also does more than inflate one class. Every strongly polarised post assigned to the neutral class is a training example removed from a minority class. At a 5.44% negative share, the smallest class is the one that this depletion reduces further.

3.3 Two weaknesses of majority voting

Pooling several annotators is a commonly proposed response when a single annotator is unreliable (Artstein and Poesio, 2008; Uma et al., 2021). The construction of MESocSentiment provides a direct test of that response: three sentiment analysers labelled every post and the majority label was assigned. The corpus skewed towards the neutral label anyway. Voting cancels error only to the extent that the voters make different errors, and the corpus authors' own reporting, summarised in Table 2, shows why that condition was not met.

Table 2. The Three Annotators Used to Build MESocSentiment, and the Labels Each Assigned (Mohd Shamsuddin et al., 2024)

Analyser	How the corpus authors describe it	Languages it is described as covering	Neutral labels assigned, of 19,714
TextBlob	"a Python library used to process textual data"	Not stated	14,305 (72.6%)
VADER	"a rule-based model … made to focus on social media texts"	Not stated	13,115 (66.5%)
pysentimiento	A toolkit "trained for sentiment analysis in English, Spanish, Portugeuse, and Italian"	English, Spanish, Portuguese, Italian	16,916 (85.8%)
Majority vote	—	—	15,422 (78.2%)

Two observations follow from Table 2. First, every analyser skewed neutral before any vote was taken, so the voting rule did not produce the skew. Second, the analyser that skewed hardest, pysentimiento at 85.8%, is the only one of the three built on a pre-trained transformer model (Devlin et al., 2019). Greater model capacity therefore did not help. None of the three analysers is described by the corpus authors as covering Malay, and one is explicitly described as trained on four other languages. What the three voters share is the language they do not cover. A

Malay affect-bearing word that one analyser does not recognise is therefore unlikely to be recognised by the others, and the vote confirms the error instead of cancelling it. This shared language gap is the first weakness of the pipeline.

The second weakness concerns the posts on which the three analysers could not agree. Where all three returned different labels there was no majority, and the authors removed those posts "to ensure the corpus only contains tweets from absolute categories". This affected 512 posts, or 2.53% of the 20,226 labelled. The difficulty is not the proportion but the composition. A post on which three analysers disagree is one where the sentiment is contested or is carried by words that none of them recognises confidently. That description fits the heavily code-switched and particle-laden posts that Section 2 identified as the hardest to annotate. The rule therefore removes the examples that a robust classifier particularly needs, and it removes them systematically rather than at random.

The effect on the corpus can be computed from the paper's own tables. Removing the contested posts left the neutral count unchanged at 15,422 while reducing the total from 20,226 to 19,714, so the neutral share rose by about two percentage points. Uma et al. (2021) question the assumption that a single correct interpretation exists for every item. Under that view, the 512 discarded posts were signal removed from the corpus rather than noise. No study examined here measures whether annotators of different types fail independently on Malaysian code-mixed text, which is the premise on which majority voting rests. The evidence above suggests that they do not fail independently when they share a language gap, and Section 5 returns to this point as a direction for future research.

4. Evaluating Annotation Reliability

Section 3 concerned how labels are produced. This section concerns how the reliability of those labels is evaluated. Reliability is normally assessed with a chance-corrected agreement coefficient, such as Cohen's κ (Cohen, 1960; Artstein and Poesio, 2008). The coefficient measures how far the agreement two annotators reach exceeds the agreement expected by chance. In computational linguistics, values above 0.8 are generally treated as good reliability, and values between 0.67 and 0.8 as allowing tentative conclusions (Artstein and Poesio, 2008).

Those conventions assume that expected chance agreement is stable across datasets, and it is not. When one category dominates the label distribution, the agreement expected by chance rises, and the same observed agreement yields a lower coefficient (Feinstein and Cicchetti, 1990; Byrt et al., 1993). Di Eugenio and Glass (2004) call this the prevalence problem. They report a case in which two coders agreeing on 90 of 100 items obtain κ values ranging from -0.048 to 0.80, depending only on how the two categories are distributed. Artstein and Poesio (2008) state the consequence for annotation directly: when one category is very common, high agreement can result from indiscriminate coding rather than from an ability to distinguish between categories.

That is the condition Malaysian code-mixed sentiment data is in. The labelling methods described in Section 3 produce heavily skewed class distributions, with the largest corpus at 78.23% neutral. The same skew that makes the labels

unrepresentative also makes the coefficient used to check them difficult to interpret, because it raises the chance agreement on which the correction depends. The difficulty runs in both directions. A low coefficient may indicate unreliable annotation, or it may indicate a dominant class. A high coefficient may indicate reliable annotation, or it may indicate indiscriminate coding. Neither interpretation can be ruled out from the coefficient alone.

Table 3 sets out what the studies introducing the four Malay-English sentiment resources report about the reliability of their annotation.

Table 3. Reliability Evidence Reported by the Four Malay-English Sentiment Corpora

Corpus	Annotators	Written guideline	Coefficient reported	What the coefficient measured
MESocSentiment (Mohd Shamsuddin et al., 2024)	None; three automatic analysers	Not applicable	None	—
MyCovid-Senti (Kong et al., 2023)	Three, independent	Yes	None	—
MY-EN-CS (Kasmuri and Basiron, 2019)	Two undergraduates proficient in Malay and English	Yes; the guideline is the paper's contribution	$\kappa = 0.83$	Whether a sentence is opinionated or factual
SentiBahasaRojak (Romadhona et al., 2022)	Five native Bahasa Rojak experts	Not stated	"Kappa" = 0.77; the variant is not specified	Whether a stock-market post is positive or negative; reported for the stock review subset only

Three observations follow. First, two of the four studies report no agreement coefficient at all, including one whose corpus was annotated by three people working from a written guideline. Second, the two coefficients that are reported were obtained for binary judgements, and neither concerns the three-class polarity task for which the resources are used. Third, Romadhona et al. (2022) report a kappa value without stating which coefficient it is. Their annotation used five experts with the final label set by majority vote, whereas Cohen's κ is defined for two annotators.

None of the four studies therefore supplies what a reliability claim requires: a coefficient of a stated type, computed for the task the labels are used for, over annotators whose number and qualification are recorded. A study that did supply them would still face the interpretation problem described above, because the class distributions these labelling methods produce are the ones that make the coefficient difficult to interpret. Reliability for this data is therefore not only unreported. With the instruments currently in use, it is not clearly achievable.

5. Synthesis: How the Challenges Compound

Sections 2 to 4 examined linguistic structure, annotation practice and reliability evaluation as three separate difficulties. Read together, they form one sequence in which each difficulty creates the conditions for the next, and the sequence returns to

its own starting point. Figure 1 sets out that sequence, and the directions for future research in Section 5.2 follow from it.

5.1 The compounding chain

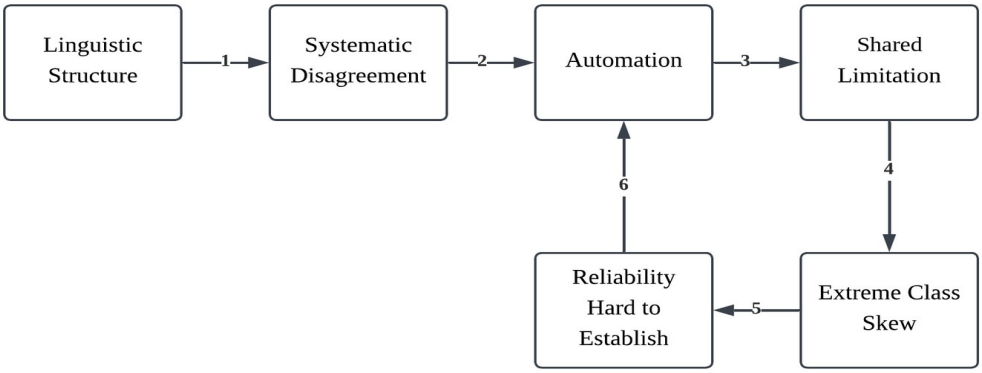


Figure 1. The Compounding Chain in Malaysian Code-Mixed Sentiment Annotation

The chain begins with the text. Because the particles that carry sentiment in Malaysian code-mixed writing can be read in more than one justifiable way, annotators may disagree systematically rather than randomly. Adding annotators reduces random disagreement, because independent errors average out. It does not reduce systematic disagreement, because each new annotator brings a stable reading of their own (Pavlick and Kwiatkowski, 2019). Competent annotation therefore requires fluent bilingual readers, and their cost and scarcity is the reason Mohd Shamsuddin et al. (2024) give for automating. The tools used for automation were designed for other languages and do not recognise Malay spellings or particles. In MESocSentiment, none of the three analysers is described as covering Malay, so pooling them confirmed a shared error instead of cancelling it. False neutrals accumulated, and the corpus acquired a class distribution that no participant intended.

That skew then undermines the check that would reveal it. A distribution skewed in that way is the condition under which a chance-corrected coefficient becomes difficult to interpret (Feinstein and Cicchetti, 1990; Byrt et al., 1993). A researcher who suspects the labels are unsound and computes a coefficient therefore cannot distinguish poor annotation from competent annotation of a skewed corpus. The defect and the inability to detect it have the same cause, so the quality check is least informative where quality is lowest. Corpora whose reliability was never established are then released, trained on and evaluated against. Manukonda and Kodali (2025) show a fine-tuned model handling the spelling variants it was shown in training. By the same reasoning, such a model cannot be expected to resolve variants it was not shown. An annotator fine-tuned on one generation of labels therefore carries their limitations into the next. That is the arrow in Figure 1 that returns to automation.

Read as a whole, the chain explains a pattern that the individual studies cannot. Each reviewed study intervenes at one point in the chain and does so competently,

but none addresses the chain as a whole. An intervention at any one point is assessed with an agreement statistic compromised at another, so the assessment cannot show whether the intervention worked. That is why competent work at each point has not accumulated into progress.

5.2 Directions for future research

The chain shows where intervention would matter, not which intervention would succeed. Four directions follow from it.

The first is agreement measurement under class skew. Prevalence-adjusted coefficients exist (Byrt et al., 1993), but no study examined here evaluates them on code-mixed sentiment data. It is therefore not known which behaves best under the distributions these corpora exhibit. The study requires no new data collection, only an existing annotated corpus, a set of candidate coefficients, and a comparison of how they behave as skew increases. It is a precondition for the other three directions.

The second is a minimum reporting standard, which follows from Table 3. Any newly released corpus should report six elements: (1) the number of annotators and their qualification; (2) the availability of the annotation guideline; (3) the variant of the coefficient and the task it was computed for; (4) per-class as well as aggregate agreement; (5) the full class distribution; and (6) the distribution of annotator judgements for each item, together with any items discarded. Per-class agreement matters because an aggregate figure on a skewed corpus is dominated by the majority class and can conceal near-total disagreement on the minority polarities. The sixth element preserves the contested cases that current pipelines delete. Uma et al. (2021) report that models trained on the full distribution of annotator judgements can outperform models trained on aggregated labels. A corpus that reports what it discarded also lets a reader judge whether its reliability was earned or obtained by removing the contested items.

The third is a shared annotation convention for affect-bearing particles such as *lah* and *kan*. Because the disagreement these particles generate is systematic, the remedy is a shared convention rather than additional annotators, and none was found among the studies reviewed here. Mohammad (2016) proposes a questionnaire that separates the speaker's emotional state, the target of the opinion and the attitude taken toward it. The questionnaire also treats sarcasm and mockery as a distinct response rather than folding them into polarity, which is the ambiguity Section 2.2 identified. Kong et al. (2023) adapt this scheme into a four-question guideline for Malaysian tweets. What is missing is its adaptation to Malaysian particles and its adoption across projects.

The fourth is to test the independence premise. Every consensus mechanism in the reviewed studies assumes that annotators of different types fail independently, and on Malaysian code-mixed text that assumption has been asserted but not measured. The study is direct: label a common sample with annotators of deliberately different competence, and measure the correlation of their errors rather than their individual accuracies. Low correlation would justify the aggregation strategies already in use. High correlation would show that consensus on this text adds confidence without adding evidence.

6. Conclusion

This review examined why reliable annotation and evaluation of Malaysian Malay-English code-mixed sentiment data have not yet been established. One common explanation concerns the model architecture, and the evidence assembled here indicates that it is incomplete. Model choice accounts for part of the difficulty, and the rest arises earlier, in the sentiment labels themselves and in the evidence offered that those labels are sound. The contribution of this review is the relation between these findings, set out in Figure 1 as a chain in which each difficulty creates the conditions for the next. An intervention at any one point in that chain is assessed by an agreement statistic compromised at another. Until reliability can be measured under the class distributions these corpora produce, no new corpus can be shown to improve on the ones examined here.

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